SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –** SIH25033
- **Problem Statement Title-** AI-Based Smart Allocation Engine for PM Internship Scheme
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**104548
- **Team Name -** CareerCrafters

CareerCrafters

IDEA TITLE

Proposed Solution- “student ↔ company matching”

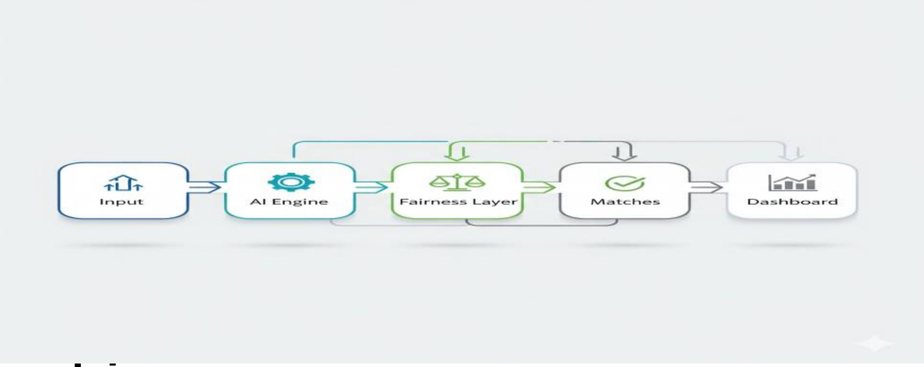
Problem Statement

- ❑ Thousands of students apply for internships under PM Scheme.
- ❑ Manual selection leads to delays, bias, and skill mismatches.
- ❑ Rural/aspirational districts & social categories often remain underrepresented.
- ❑ Need for a scalable, automated, and fairness-driven solution.

Many Applicants

OBJECTIVE AND PROPOSED SOLUTION

- **<u>Objective</u>:** Build an AI/ML-based smart system to match candidates with opportunities.



- **Solution Approach:**
- Candidate profile + internship data as input.
- AI/ML engine performs similarity scoring & ranking.
- Capacity & affirmative-action constraints ensure fairness.
- Frontend prototype demonstrates end-to-end workflow

CareerCrafters

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025

- Built with React and TypeScript(UI/UX components)
- Fast development powered by Vite
- Backend logic and data managed with Convex (TypeScript-based serverless functions) and Python.
- Node.js ecosystem for scripts and package management
- Continuous integration via TypeScript Compiler and npm tools
- Styled using CSS modules and global styles

CareerCrafters

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025

IMPACT

- **For Industry/Recruiter:** Access to pre-screened, best-fit candidates with reduced bias and efficient hiring.
- **For Students:** Fair and personalized career opportunities through AI-driven role matching, mentorship, and guidance.
- **Administration/Management:** Transparent, fair, and data-driven placement processes that enhance institutional reputation and growth.

BENEFITS

1. Students get fair, bias-free access to opportunities.
2. Recruiters save time and resources with better candidate matches.
3. Strengthens trust among students, alumni, and recruiters.
4. Enhances credibility of the institution through transparent processes.

CareerCrafters

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

1.Artificial Intelligence in Recruitment & Internship in Matching

- NLP and embeddings for skill-based candidate-job matching.
- S. Gupta et al., “AI alent Acquisition,” IJCA, 2021.

2. Machine Learning Models for Ranking & Recommendation

- XGBoost/LightGBM for candidate-job fit.
- J. Chen, “Ranking Algorithms for Recruitment Systems,” IEEE, 2020.

3. Fairness & Affirmative Action in Automated Systems

- Representation of marginalized groups in allocation
- B. Mehrabi et al., “Fairness in Machine Learning,” ACM CS, 2019

4. Constraint-Based Allocation Algorithms

- Linear programming / greedy optimization
- V. Kumar, “Optimization Methods for Resource Allocation,” Springer, 2018